AFAQ ELEVATORS

PROJECT DOCUMENTATION

Project Planning • Stakeholder Analysis • Database Design • UI/UX Design

Data Analytics Project | Python • SQL • Power BI

Based on the supplied AFAQ workbook, dataset guide, project roadmap and dashboard screenshot.

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1. Executive Summary

AFAQ Elevators is an elevator installation and maintenance business case used to demonstrate how operational, customer, project, maintenance, supply-chain, and finance data can be organized and analyzed. The project brings the supplied Excel workbook into a structured analytics workflow using Python, SQL, and Power BI.

The supplied workbook contains 15 worksheets and 514,905 data records (excluding header rows). The documented workflow covers data understanding and quality review, schema design, SQL analysis, dashboard reporting, validation, and presentation.

2. Project Overview

2.1 Business context

The dataset guide describes AFAQ Elevators as a Saudi-based hypothetical business model covering elevator installation and maintenance, contracts, spare parts, and financial collection. The data links customers and elevators to projects and service activity, then to purchasing, invoicing, and payments.

2.2 Problem statement

Operational and financial information is distributed across multiple related tables. Without a consistent data model and reporting layer, management may find it difficult to monitor service activity, fault patterns, project delivery, parts availability, supplier performance, and receivables in one place.

2.3 Project objectives

- Understand the structure and quality of the supplied data.
- Design and validate a relational database model connecting the 15 subject areas.
- Use SQL to combine data and answer business questions.
- Use Python for profiling, cleaning support, and deeper analysis.
- Present operational and management indicators through a Power BI dashboard.
- Document assumptions, data-quality issues, metric definitions, and implementation evidence.

2.4 Scope

In scope: data profiling and cleaning review; database/schema design; relationships among the supplied tables; analytical SQL; Python analysis; Power BI reporting; project documentation and demonstration.

Out of scope unless separately implemented: real-time elevator telemetry, live dispatch/control of physical elevators, predictive maintenance in production, and integration with external enterprise systems. These capabilities must not be claimed based only on the supplied historical workbook or dashboard screenshot.

3. Project Planning

3.1 Delivery approach

The project follows the sequence Python → SQL → Power BI, with iterative feedback between phases. Findings during SQL development or dashboard testing may require returning to Python to correct data preparation or revise the model.

3.2 Work plan

Phase	Main activities	Expected output
1. Discover	Inspect worksheets, fields, data types, row counts, date ranges, nulls, duplicates and quality flags.	Data profile and issue log
2. Prepare	Define cleaning rules, standardize types, review missing/invalid values, retain raw	Documented clean datasets

	copies.	
3. Model	Identify candidate keys, test uniqueness and foreign-key matches, define cardinality.	Validated schema and ERD
4. SQL	Create relational tables, load prepared data, write joins and analytical queries.	SQL scripts and query outputs
5. Analyze	Explore maintenance, faults, projects, parts, suppliers, invoices and payments.	Analysis results and findings
6. Power BI	Build model, measures, report pages, filters and visual interactions.	Power BI report
7. Validate	Reconcile totals, test measures and filters, inspect exceptions and document limitations.	QA checklist and corrections
8. Present	Finalize documentation, presentation, repository and recorded walkthrough.	Submission package

3.3 Proposed dashboard pages

- Operations Overview: branch activity, elevator asset counts, service activity and project status.
- Maintenance & Faults: fault volume, severity, downtime, unresolved cases, repeat faults and technician workload.
- Finance: invoiced revenue, direct costs where reliable, net payments, outstanding balances and aging.
- Supply Chain: supplier delivery performance, purchase orders, parts usage, inventory movements and shortages.

3.4 Project risks and mitigations

Risk	Potential impact	Mitigation
Unvalidated keys or relationships	Incorrect joins and duplicated totals	Test key uniqueness, orphan records and row multiplication before finalizing ERD
Missing or inconsistent values	Misleading KPIs	Document cleaning rules and expose exclusions
Snapshot / future-dated records	Distorted period reporting	Confirm dataset snapshot date and review QF_ flags
Financial refunds / payment statuses	Incorrect paid and outstanding amounts	Use signed payment amounts and reconcile invoice-level balances
Dashboard values not tied to workbook	Unsupported claims in report	Reconcile screenshot metrics to Power BI model and workbook before sign-off

4. Stakeholder Analysis

Stakeholders are grouped by the decisions they need to make and the information the project should provide.

Stakeholder	Information needs	Project response / value
Management / branch managers	Branch activity, service quality, project progress and revenue	Operations overview and branch comparisons
Maintenance supervisors	Faults, severity, downtime, repeat faults and technician workload	Maintenance & Faults analysis
Finance team	Revenue, payments, overdue invoices and outstanding balances	Finance reporting and receivables aging
Procurement / inventory team	Supplier delivery, part usage, purchase orders and stock movement	Supply Chain analysis
Technicians	Visit history, fault assignment and repair performance	Service history and workload measures
Project evaluator	Traceable methods, coherent data model and reproducible outputs	Documentation, code repository, dashboard and demo

4.1 Communication and review

- Confirm business definitions and KPI formulas with the project team before publishing results.
- Review ERD relationships with the SQL implementation owner.
- Review dashboard measures with the Power BI owner.
- Keep a shared issue log for data-quality exceptions, assumptions and pending checks.

5. Dataset Description & Data Inventory

The following counts were read from the supplied Excel workbook. Each count excludes the header row. Total: 514,905 records across 15 worksheets.

Worksheet	Records	Purpose
Branches	10	Branch and region reference
Customers	10,000	Customer and account attributes
Employees	420	Employee and technician records
Elevators	12,000	Elevator asset register
Contracts	8,500	Customer/elevator service contracts
Projects	4,000	Installation and upgrade projects
MaintenanceVisits	160,000	Planned and completed service visits
Faults	35,000	Fault, severity and restoration records
Suppliers	75	Supplier directory
Parts	900	Spare-parts catalogue
PurchaseOrders	9,000	Parts purchasing and delivery
InventoryTransactions	70,000	Inventory movement ledger
PartsUsage	75,000	Parts consumed during service
Invoices	60,000	Customer billing and cost records
Payments	70,000	Invoice payments and refunds

The workbook includes QF_ fields that appear to be data-quality flags and AN_ fields that appear to be analysis/helper fields. Their exact business logic should be checked against the data-generation rules or confirmed by the project team before they are used in published metrics.

6. Database Design

6.1 Design approach

The proposed relational model uses the ID columns as candidate primary keys and the matching ID columns in related tables as candidate foreign keys. The candidate model is included as a separate Mermaid ERD file in the submission package. It is a design proposal, not proof that every relationship has already been enforced in SQL.

6.2 Candidate primary keys

Table	Candidate primary key
Branches	BranchID
Customers	CustomerID
Employees	EmployeeID
Elevators	ElevatorID
Contracts	ContractID
Projects	ProjectID
MaintenanceVisits	VisitID
Faults	FaultID
Suppliers	SupplierID
Parts	PartID
PurchaseOrders	PurchaseOrderID
InventoryTransactions	TransactionID
PartsUsage	PartsUsageID
Invoices	InvoiceID
Payments	PaymentID

6.3 Main candidate relationships

- Branches → Customers, Employees, Elevators, Projects, PurchaseOrders, InventoryTransactions, and Invoices.
- Customers → Elevators, Contracts, Projects, MaintenanceVisits, Faults, Invoices, and Payments.

- Elevators → Contracts, Projects, MaintenanceVisits, Faults, and PartsUsage.
- Employees → MaintenanceVisits, Faults, and PartsUsage (technician references).
- Contracts → MaintenanceVisits and Invoices.
- MaintenanceVisits → Faults, InventoryTransactions, PartsUsage, and Invoices.
- Faults → InventoryTransactions and PartsUsage.
- Suppliers → Parts and PurchaseOrders.
- Parts → PurchaseOrders, InventoryTransactions, and PartsUsage.
- Projects → Invoices.
- Invoices → Payments.

6.4 Validation required before final ERD

- Confirm each candidate PK is unique and non-null.
- Check every candidate FK against its parent key and count unmatched (orphan) values.
- Check whether nullable relationships are optional and whether the data supports one-to-many cardinality.
- Check whether purchase orders are truly one row per part/order or whether a separate order-header / order-line model is needed.
- Avoid enforcing constraints until data exceptions have been reviewed.
- Reconcile the ERD against the actual SQL CREATE TABLE statements and Power BI relationships.

7. Data Quality & Preparation

7.1 Recommended checks

- Missing values in key and descriptive fields.
- Duplicate candidate primary keys and suspected duplicate business records.
- Invalid or inconsistent date values, including dates after the dataset snapshot or current date.
- Orphan foreign-key values across all candidate relationships.
- Negative or implausible quantities, costs, durations and balances.
- Payment refunds and signed payment amounts.
- Inventory transaction sign logic and negative stock balances.
- Fault and maintenance chronology, technician hire dates, and elevator commissioning dates.

7.2 Quality flags

QF_ fields should be treated as exception indicators until their definitions are verified. Examples in the workbook include flags for records after snapshot/today, duplicate-looking records, mismatched technician references, negative inventory balances, and financial status differences.

7.3 Data preparation principles

- Keep an unchanged raw copy of every source worksheet.
- Apply transformations in reproducible Python or SQL scripts.
- Record each cleaning decision, affected row count and reason.
- Do not silently delete records flagged by QF_ fields.
- Keep a separate exception report for unresolved issues.

8. KPI Framework & Calculation Definitions

The following are recommended definitions for the project. They should be matched to the exact columns, filters and DAX / SQL implementation used by the team before being labelled as final.

KPI	Suggested calculation	Notes / validation
Total Elevators	Distinct count of ElevatorID	Define whether retired/out-of-service assets are included
Daily Trips	Count of trip records for the selected date	The supplied workbook inventory does not list a Trips worksheet; confirm the source used by the dashboard
Average Wait Time	Average of recorded wait duration	A wait-time field is not visible in the supplied workbook field inventory; confirm source
Maintenance Due	Count of elevators with next maintenance date within the next 7 days	Confirm that a due-date field exists in the dashboard data source
Elevator Status %	Elevators in each status ÷ total elevators × 100	Use a consistent status mapping and denominator
Fault Rate	Fault count ÷ elevator count (or per 100 elevators)	State the period and denominator
Average Downtime	Average DowntimeHours for the selected faults	Decide whether open faults are included
Project Cost Variance	ActualCostSAR − BudgetSAR	Positive values indicate over budget
Delivery Delay	ActualDeliveryDate − ExpectedDeliveryDate	Define treatment of open orders and early deliveries
Outstanding Balance	Invoice amount − net signed payments	Reconcile refunds, voids and status flags

8.1 Dashboard screenshot values (as displayed)

Displayed item	Screenshot value
Total Elevators	8
Total Trips (Today)	1,248
Average Wait Time	24 sec
Maintenance Due	1 within 7 days
Elevator status	6 In Service (75%); 1 Maintenance (12%); 1 Out of Service (12%)

9. UI/UX Design & Dashboard

The supplied dashboard screenshot uses a dark navy interface with a left navigation rail and an operations overview. The visible navigation includes Overview, Elevators, Maintenance, and Reports.

9.1 Main visual components

- KPI cards: Total Elevators, Total Trips (Today), Average Wait Time, and Maintenance Due.
- Time-series line chart: Trips Throughout the Day.
- Doughnut chart: Elevator Status distribution.
- Elevator detail table: ID, status, current floor, direction and trips today.

9.2 UX observations

- The top KPI cards provide a quick operational summary.
- The status chart uses distinct colors and a legend to communicate service availability.
- The trend chart supports time-of-day comparison.
- The detailed table supports drill-down from summary to individual elevator records.
- For accessibility, retain readable contrast, meaningful chart titles, clear units, and non-color status labels.

9.3 Final dashboard screenshot



Figure 1. Elevator Management System dashboard screenshot supplied by the project team (30 October 2026).

10. Analytical Questions & Expected Outputs

Business question	Relevant data areas	Potential output
Which branches have the highest fault volume?	Branches, Elevators, Faults	Fault count by branch and period
How do downtime and resolution vary by severity?	Faults	Downtime and resolution status by severity
Which projects are late or over budget?	Projects	Schedule variance and cost variance
Which suppliers have delivery delays?	Suppliers, PurchaseOrders	Average delay and late-order rate
Which parts are used most often?	Parts, PartsUsage, InventoryTransactions	Usage quantity and stock movement
How much is invoiced, paid and outstanding?	Invoices, Payments	Invoice, net payment and receivable analysis
Which data-quality issues occur most often?	QF_ and AN_ fields across tables	Exception counts and quality scorecard

Actual findings should be inserted only after the team has run and checked the corresponding SQL/Python analysis. Do not replace this section with invented rankings or unsupported percentages.

11. Testing, QA & Acceptance Criteria

- All 15 workbook worksheets are accounted for in the data inventory.
- Row counts before and after loading are reconciled.
- Candidate primary keys are unique and non-null, or exceptions are documented.
- Candidate foreign-key orphan counts are reviewed.
- SQL query totals agree with independently checked calculations.

- Power BI measures are checked against sample records and SQL outputs.
- Filters, slicers, date ranges and cross-highlighting behave as intended.
- Screenshot KPI values are reconciled to the report data model.
- Documentation, repository, presentation and demo describe the same final implementation.

11.1 Known open verification items

- Confirm dashboard source for trip count, average wait time, current floor and direction.
- Confirm why the dashboard shows 8 elevators while the workbook contains 12,000.
- Validate all candidate ERD relationships against the actual SQL schema.
- Add final, evidence-backed analysis findings and KPI definitions from the team's implementation.

12. Deliverables & Submission Structure

Deliverable	Recommended location / format	Status
Project Documentation	Google Drive: final PDF and editable DOCX	Prepared in this document; review verification notes
Project Planning	Inside Documentation	Included
Stakeholder Analysis	Inside Documentation	Included
Database Design / ERD	Inside Documentation + Mermaid source in repository	Candidate; validate against SQL
UI/UX Design	Inside Documentation with dashboard screenshot	Screenshot included
SQL source code	GitHub repository / SQL folder	Provided by project team
Python source code	GitHub repository / Python folder	Provided by project team
Power BI report	GitHub repository / PowerBI folder (.pbix)	Provided by project team
Presentation	Google Drive PDF / PPTX	Separate presentation file
Video demonstration	Google Drive video link	Record and upload by team

12.1 Suggested GitHub structure

```
AFAQ-Elevators-Project/
├── SQL/
│   └── database_script.sql
├── Python/
│   └── analysis.py
├── PowerBI/
│   └── AFAQ_Elevators_Dashboard.pbix
├── Documentation/
│   ├── AFAQ_Elevators_Final_Documentation.pdf
│   ├── AFAQ_Elevators_Final_Documentation.docx
│   └── ERD_candidate.mmd
└── README.md
```

Final submission links to provide: (1) Documentation folder/file on Google Drive, (2) GitHub repository URL for source code, and (3) Presentation PDF plus recorded demonstration video on Google Drive. Set link permissions according to the instructor's requirements.

13. Conclusion

The AFAQ Elevators project provides an end-to-end analytics workflow across operations, maintenance, projects, supply chain and finance. The documentation establishes the project objectives, stakeholders, delivery plan, dataset inventory, candidate data model, quality checks, KPI definitions and dashboard interface.

The project should be considered ready for final submission only after the team validates the candidate relationships, reconciles the dashboard metrics with their actual data source, inserts verified analytical findings, and checks that the GitHub repository and demonstration video are accessible.